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Construction of the prediction model and analysis of key winning factors in world women's volleyball using gradient boosting decision tree

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This study aims to analyze the key factors contributing to victories in world women's volleyball matches and predict match win rates using machine learning algorithms. Initially, Grey Relational Analysis (GRA) was employed to analyze the fundamental match data of the top six teams over three major world tournaments during the 2020 Olympic cycle (a total of 142 matches, 505 sets, and 27 metrics). The 27 metrics were used as subsequences, and the set win rate served as the parent sequence to identify metrics with a high contribution to match victories. Subsequently, the Gradient Boosting Decision Tree (GBDT) algorithm was utilized to construct a prediction model for match win rates, using the selected metrics as input features and set win rates as output features. The input metrics were ranked by their contribution to determine the most influential factors on match victories. The results indicate that spike scoring rate, blocking height, excellent defense rate, serve scoring rate, block scoring rate, proportion of serve scores, and proportion of block scores significantly impact match victories. Among these, spike scoring rate and blocking height are decisive, with feature importance values of 0.45 and 0.3, respectively. The constructed GBDT model demonstrated good predictive performance, capable of predicting match win rates. The model parameters are as follows: learning rate (learning-rate) of 0.1, number of trees (n-estimators) of 150, and maximum depth of the tree model (max-depth) of 2. The model's accuracy metrics on the test set are: MSE = 0.002, MAE = 0.0322, R^2 = 0.8497, and MAPE = 4.77%. The average relative error of the model validation is 5.30%, with R^2 = 0.743. This study not only identifies the key factors contributing to victories in world women's volleyball but also demonstrates the innovative application of combining Grey Relational Analysis with the Gradient Boosting Decision Tree algorithm in the volleyball domain. The findings provide data-driven insights to support coaches in training design and in-game decision-making, highlighting important practical value.

Keywords World women's volleyball, GBDT, Gradient boosting decision tree, Winning factors, Prediction model

Machine learning (ML), as a key branch of artificial intelligence, can identify patterns and extract regularities through iterative learning from large-scale data, thereby enabling robust prediction and scientific decision-making. It has been widely applied in fields such as medicine[1,2], biology[3], and management[4]. In recent years, its applications in sports science have expanded considerably, making it an essential tool for driving the digital transformation of sports. Specifically, the research and practical pathways of ML in sports can be outlined across several dimensions. First, in match outcome prediction and performance analysis, ML methods transcend the limitations of traditional regression and descriptive statistics, allowing for more effective exploration of nonlinear relationships among multidimensional indicators[5,6]. Second, in training load monitoring and performance evaluation, data collected from wearable devices and sensors can be leveraged by ML models to achieve precise monitoring and dynamic prediction of training volume, intensity, and skill level[7,8]. Third, in the domain of injury risk identification and prevention, ML combined with biomechanical and medical monitoring indicators can detect potential risks in advance and provide personalized intervention strategies[9–13]. Fourth, in talent

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identification and long-term athlete development, ML models based on multimodal data can quantitatively assess athletic potential, thereby supporting evidence-based selection and training planning[14–16]. Overall, ML has gradually become an important support tool in sports science research and practice, demonstrating broad prospects and value in data-driven scientific decision-making, refined training, and health protection.

The Gradient Boosting Decision Tree (GBDT) algorithm, proposed by Friedman in 2001[17], is a type of ensemble learning algorithm suitable for handling regression and classification problems. It constructs a series of weak learners (i.e., decision trees) and aggregates their results to form a strong learner, thus enabling the prediction of target variables. In recent years, GBDT has been widely applied across various fields. This algorithm exhibits strong computational capabilities for different types of data, including continuous variables, discrete variables, and missing values. Especially when dealing with complex data structures and capturing non-linear relationships, GBDT iteratively trains multiple tree models to explore hidden patterns and relationships between variables, providing highly accurate predictions. Additionally, due to its ensemble learning nature, GBDT typically achieves better predictive performance than a single decision tree. This algorithm not only shows robustness to outliers but also allows for controlling overfitting by adjusting hyperparameters such as learning rate, the number of trees, and maximum depth, thereby enhancing the model's generalization ability. GBDT is valuable for performance prediction and plays a significant role in feature selection, data understanding, and decision-making. It has achieved notable success in fields such as materials science[18], healthcare[19], transportation[20], chemistry [21], power[22], and environmental science[23]. In the sports domain, research has applied GBDT to predict football player values and match outcomes[24,25], as well as predicting college students' sports interests[26] and psychological health levels[27], all with positive results.

At present, the competitive level of world women's volleyball has improved markedly, with matches characterized by faster pace, longer rallies, and greater diversity of play patterns. This has increased the uncertainty of outcomes, making the objective identification of winning factors increasingly critical. Although previous studies have made some progress in analyzing technical–tactical indicators and predicting match outcomes, several limitations remain. (1) Sample sizes are insufficient, with most research focusing on a single event or a limited number of teams, lacking systematic datasets across multiple cycles and competitions[28]; (2) the coverage of indicators is limited, often emphasizing a single position or a small subset of technical–tactical variables, which fails to comprehensively reflect the overall contributions of multidimensional factors such as attack, block, serve, setting, and defense[29,30]; (3) methodological interpretability is inadequate, as prior studies have mainly relied on regression analysis or descriptive statistics[31], which can reveal partial correlations but are unable to capture nonlinear mechanisms among technical–tactical indicators. Meanwhile, although machine learning methods have been introduced into volleyball research, they have primarily been applied in areas such as motion analysis[32–34], technical–tactical analysis[35–37], physical fitness assessment[38,39], training load modeling[40], and performance evaluation[41,42]; however, research on match outcome prediction and the elucidation of decisive factors remains limited, with a lack of systematic investigation into the multidimensional mechanisms underlying winning determinants[33,36,37].

Building on this, the present study employs a combined modeling approach of Grey Relational Analysis (GRA) and Gradient Boosting Decision Trees (GBDT), using match data from the three major tournaments of the 2020 Olympic cycle and incorporating a multidimensional systematic indicator framework. GRA was first applied for feature reduction; compared with traditional regression analysis or principal component analysis, GRA is better suited to contexts with limited samples and numerous indicators, as it directly measures the correlation strength between technical–tactical metrics and win rates, thereby reducing redundancy and enhancing both modeling efficiency and interpretability. Subsequently, GBDT was introduced to model the selected features. Compared with neural networks, support vector machines, or random forests, GBDT is particularly appropriate for volleyball technical–tactical data characterized by moderate sample sizes, diverse variable types, and complex relationships. It not only robustly captures nonlinear associations while mitigating the risk of overfitting but also outputs interpretable results through feature importance ranking, thereby achieving a balance between predictive performance and practical applicability. The findings provide methodological support for the scientific preparation of world-class women's volleyball teams and offer insights for extending the application of machine learning methods to other competitive sports.

Data and methods
Data preparation
In machine learning modeling, sample size and indicator dimensionality are critical conditions for ensuring accuracy and robustness of results. This study draws on match data from the top six teams across the three major world tournaments of the Tokyo 2020 Olympic cycle—2018 World Cup (59 matches, 211 sets), 2019 World Championship (51 matches, 181 sets), and 2020 Olympic Games (32 matches, 113 sets)—amounting to a total of 142 matches and 505 sets. Although this scale may not reach the level of "big data" in the general sense, it represents a relatively comprehensive and well-structured dataset within the domain of volleyball, capable of systematically reflecting the statistical characteristics of elite competitions, thereby demonstrating strong representativeness and research value. In addition, this study constructed a multidimensional dataset encompassing both athletes' morphological characteristics (age, height, weight, BMI, spike height, block height, etc.) and technical–tactical statistics (across six dimensions: attack, block, serve, setting, defense, and reception). This systematic framework provided solid support for feature selection and model construction, enabling the in-depth identification of decisive factors and robust modeling even under conditions of limited sample size.

The data set includes a total of 10 teams. The United States team ranked in the top six in all three tournaments and won the championship at the 2020 Olympics. China, Serbia, Italy, Korea, Japan, and Brazil each ranked in the top six in two tournaments, with Serbia and China winning the 2018 World Championship and the 2019 World Cup, respectively. The Netherlands, Turkey, and Russia each made it to the top six once. The set win rates (number of sets won/total sets played) and the final rankings for each team are detailed in Table 1. Using the Data Volley volleyball statistics software and relevant data published on the International Volleyball Federation's official website, we identified 27 fundamental metrics: age (X_1), height (X_2), weight (X_3), spike height (X_4), block height (X_5), BMI (X_6), attack score ratio (X_7), block score ratio (X_8), serve score ratio (X_9), spike success rate (X_{10}), spike error rate (errors/total) (X_{11}), spike general rate (effective/total) (X_{12}), block success rate (X_{13}), block error rate (X_{14}), block general rate (X_{15}), serve success rate (X_{16}), serve error rate (X_{17}), serve general rate (X_{18}), excellent set rate (X_{19}), set error rate (X_{20}), set general rate (X_{21}), excellent defense rate (X_{22}), defense error rate (X_{23}), defense general rate (X_{24}), excellent reception rate (X_{25}), reception error rate (X_{26}), and reception general rate (X_{27}). These 27 metrics and the set win rate constitute the base data set for Grey Relational Analysis (GRA) calculations, aiming to identify the primary indicators significantly influencing match victories.Prior to model construction, the raw data were systematically preprocessed. First, for variables with a low proportion of missing values, mean imputation was applied, and subsequent tests confirmed that this had a negligible effect on the overall distribution. All feature variables were then standardized using Z-scores to ensure comparability of scale and reduce statistical bias. Since the output variable was the continuous set win rate, issues of class imbalance common in classification tasks did not arise. However, stratified sampling based on the distribution of set win rates was employed for splitting the training and test sets, thereby ensuring consistency of the overall distribution and reducing the potential impact of partitioning bias.

Research methods

Grey relational analysis

Grey Relational Analysis (GRA) is a statistical technique used to quantify the degree of association between multiple factors within a system and a target variable. It enables the identification of the relative importance of different indicators to the target variable, thereby facilitating dimensionality reduction and structural optimization of the feature space[43]. Compared with traditional dimensionality-reduction methods such as regression analysis or principal component analysis, GRA does not rely on assumptions of linearity or specific data distributions. It is particularly suitable for research contexts characterized by limited sample sizes and numerous variables, and provides robust assessments of association strength when nonlinear relationships among variables are pronounced[44].In this study, GRA was employed as an exploratory and structural optimization tool to measure the univariate association strength between each technical-tactical feature and the set win rate. Based on the calculated association degrees, lightweight dimensionality reduction was performed to minimize redundancy, enhance the signal-to-noise ratio, and optimize feature structure, thereby improving the training efficiency and stability of the subsequent GBDT model. The analytical procedure involved constructing a reference sequence of set win rates and comparative sequences for each technical-tactical indicator, calculating grey relational coefficients, and deriving the overall grey relational degree. Indicators were then ranked by their relational degree, and those with higher correlations were selected as input features for the GBDT model.To validate the rationality of this procedure, two modeling schemes—"full-feature modeling" and "GRA-optimized modeling"—were compared. The results showed that the GRA-optimized feature set outperformed the full-feature scheme across all performance metrics, including R^2, MAE, and MSE, indicating that under the data

Tournament	Country	Number of matches	Number of sets	Sets won	Sets lost	Set win rate	Ranking
2018 World Championship	SRB	13	44	34	10	0.773	1
	ITA	13	49	36	13	0.735	2
	CHN	13	47	36	11	0.766	3
	NED	13	48	31	17	0.646	4
	USA	12	47	29	18	0.617	5
	JPN	12	46	28	18	0.609	6
2019 World Cup	CHN	11	36	33	3	0.917	1
	USA	11	40	30	10	0.750	2
	RUS	11	41	27	14	0.659	3
	BRA	11	40	24	16	0.600	4
	JPA	11	42	23	19	0.548	5
	KOR	11	40	21	19	0.525	6
2020 Olympics	USA	8	28	21	7	0.750	1
	BRA	8	28	21	7	0.750	2
	SRB	8	25	19	6	0.760	3
	KOR	8	30	12	18	0.400	4
	TUR	6	25	14	11	0.560	5
	ITA	6	21	11	10	0.524	6
Total	–	186	677	450	227	–	–

Table 1. Overview of subjects investigated in the women's volleyball match winning analysis. The number of matches in the table is calculated based on the team unit, and the same match may be counted separately by both teams, resulting in a discrepancy with the total number of matches mentioned in the text.

conditions of this study, the GRA-based optimization process improved the model's generalization ability and stability. Therefore, GRA in this research was positioned as an auxiliary step for feature correlation assessment and structural optimization, designed to provide the GBDT model with a more informative and interpretable set of inputs, ultimately achieving synergistic enhancement in both predictive performance and model transparency. The detailed computational steps are as follows:

Step 1: Establishing Analysis Sequences

Selecting the Reference Sequence (Parent Sequence): The match win rates of the top six teams from the three major tournaments during the 2020 Olympic cycle (2020 Olympics, 2019 Women's Volleyball World Cup, 2018 Women's Volleyball World Championship) serve as the reference sequence. The match win rate (P) is calculated as P = x/y(where x is the number of sets won, and y is the total number of sets played).

Establishing the Comparison Sequences (Sub-sequences): The fundamental metrics of the top six teams are used as comparison sequences. These metrics include age (X_1), height (X_2), and so on up to

$$\begin{gathered} X_{27}\,(\text{general reception rate})\,. \\ \text{The formula}:\, X_i = X_i\,(k)\,.\; k=1,2,\ldots,18;\; i=1,2,\ldots,27. \end{gathered} \tag{1}$$

Steps 2–6: These include dimensionless processing, calculating the grey relational coefficient, and determining the grey relational grade.

The formula for calculating the grey relational grade is as follows:

$$\gamma\left(X_0, X_i\right) = \frac{1}{n}\sum_{k=1}^{n}\gamma X_0\left(k\right) - X_i\left(k\right); i=0,1,2,\ldots\ldots 27 \tag{2}$$

Gradient boosting decision tree

The Gradient Boosting Decision Tree (GBDT) algorithm is an efficient machine learning technique based on the boosting method. It constructs a strong predictor by iteratively combining multiple decision tree models. In each iteration, GBDT focuses on the residuals of the previous prediction (the differences between the actual observed values and the current model's predicted values). These residuals are used as the new target variable to train the next decision tree, thereby improving the prediction accuracy through the iteration of several decision trees.

In this study, the Gradient Boosting Decision Tree (GBDT) algorithm was selected as the core modeling method, primarily based on the structural characteristics of volleyball match data and the alignment with the research objectives. Volleyball match indicators include continuous variables and exhibit complex nonlinear and interactive relationships among them. GBDT iteratively integrates weak learners (decision trees) to fit residuals, effectively capturing high-order nonlinear features while maintaining high predictive accuracy and model stability through the regulation of hyperparameters such as learning rate and tree depth[45]. Compared with other algorithms, GBDT not only ensures predictive performance but also provides a stable ranking of feature importance, thereby enabling the identification and interpretation of key winning factors in line with this study's dual objectives of predictive accuracy and interpretability. In addition, GBDT demonstrates strong robustness to missing values and outliers, making it suitable for handling inevitable data fluctuations in real match environments[46]. During the model selection stage, systematic comparisons were conducted with mainstream algorithms, including Random Forest (RF), Support Vector Machine (SVM), and Neural Network (NN) models. The results showed that GBDT outperformed the other models in predictive accuracy, feature importance consistency, and model stability. RF exhibited greater variability in feature weight distribution, SVM was highly sensitive to parameter tuning and lacked interpretability[47], and NN tended to overfit under small- and medium-sample conditions[48]. Integrating theoretical suitability with empirical comparison results, GBDT was ultimately determined to be the core modeling method of this study. Specifically, the computational steps are as follows:

Step 1: Initialize the model

Initialize the model's prediction, which can be chosen as the average of the training targets:

$$\mathrm{F0x} = \frac{1}{N}\sum_{i=1}^{N} y_i \tag{3}$$

Step 2: Compute Residuals

Calculate the residuals between the current model's predictions and the actual target values:

$$\mathrm{r_{ij}} = \mathrm{y_j} - \mathrm{F_i} - 1\left(\mathrm{x_j}\right) \tag{4}$$

Step 3: Train Decision Tree

Use the residuals as the target to train a decision tree model to fit the residuals. Typically, mean squared error (MSE) is used as the splitting criterion to find the split points that minimize the residuals.

Step 4: Compute Decision Tree Predictions

For the i decision tree, calculate its prediction for sample j:

$$\mathrm{h_i}\left(\mathrm{x_j}\right) = \mathrm{Tree}(\mathrm{x_j};\theta_i) \tag{5}$$

Step 5: Update the Model

Update the model by adding the weighted predictions of the newly trained decision tree to the current model's predictions:

$$\mathrm{F_i\left(x_j\right) = F_i - 1\left(x_j\right) + learning - rate * h_i\left(x_j\right)} \tag{6}$$

Step 6: Update Residuals
Use the new model predictions to calculate the new residuals:

$$\mathrm{R_{ij} = r_{ij} - learning\_rate * h_i\left(x_j\right)} \tag{7}$$

Step 7: Iterate
Repeat steps 3 to 6, iteratively training multiple decision trees, each time further reducing the residuals.
Step 8: Final Model Prediction
The final model prediction is the cumulative sum of the predictions of all the decision trees, each weighted by the learning rate:

$$F_{final}(x_j) = F_0(x_j) + \sum_{(i=1)}^{M} learing - rate * hi(xj) \tag{8}$$

Research results

Analysis of grey relational analysis results

Based on the foundational data, indicators were screened to select those with high correlation to set win rates. The grey relational analysis (GRA) provided the relational degrees for each indicator, as shown in Table 2. The average relational degree was found to be 0.691. Using the method of defining key indicators based on the average relational degree, 13 indicators with a relational degree greater than the average (the critical point) were selected as key indicators: X10 (spike success rate, i.e., spikes/total attempts), X_{13} (block success rate), X_5 (block height), X_{16} (serve success rate), X_9 (serve score ratio), X_{22} (excellent defense rate), X_2 (height), X_{18} (general serve rate), X_{19} (excellent set rate), X_8 (block score ratio), X_{21} (general set rate), X_{15} (general block rate), and X_1 (age). This approach effectively eliminates indicators with lower relevance to match victories while retaining a substantial amount of data information. The relational degrees between the indicators are shown in Fig. 1.

Evaluation Indicator	Grey Relational Degree	Weight	Rank
X_{10}(Spike success rate)	0.851	0.046	1
X_{13} (Block success rate)	0.800	0.043	2
X_5 (Block height)	0.793	0.043	3
X_{16} (Serve success rate)	0.753	0.040	4
X_9(Serve score ratio)	0.752	0.040	5
X_{22} (Excellent defense rate)	0.746	0.040	6
X_2 (Height)	0.732	0.039	7
X_{18} (General serve rate)	0.723	0.039	8
X_{19} (Excellent set rate)	0.722	0.039	9
X_8(Block score ratio)	0.708	0.038	10
X_{21} (General set rate)	0.706	0.038	11
X_{15} (General block rate)	0.697	0.037	12
X_1 (Age)	0.693	0.037	13
X_{27} (General reception rate)	0.689	0.037	14
X_3(Weight)	0.684	0.037	15
X_4 (Spike height)	0.684	0.037	16
X_{20} (Set error rate)	0.665	0.036	17
X_{11} (Spike error rate)	0.643	0.034	18
X_{25} (Excellent reception rate)	0.643	0.034	19
X_{12}(General spike rate)	0.642	0.034	20
X_{24} (General defense rate)	0.631	0.034	21
X_6 (BMI)	0.629	0.034	22
X_{14} (Block error rate)	0.626	0.034	23
X_{26} (Reception error rate)	0.623	0.033	24
X_7 (Spike score ratio)	0.610	0.033	25
X_{17} (Serve error rate)	0.607	0.033	26
X_{23} (Defense error rate)	0.607	0.033	27

Table 2. Ranking of match win rate and grey relational degrees for various indicators.

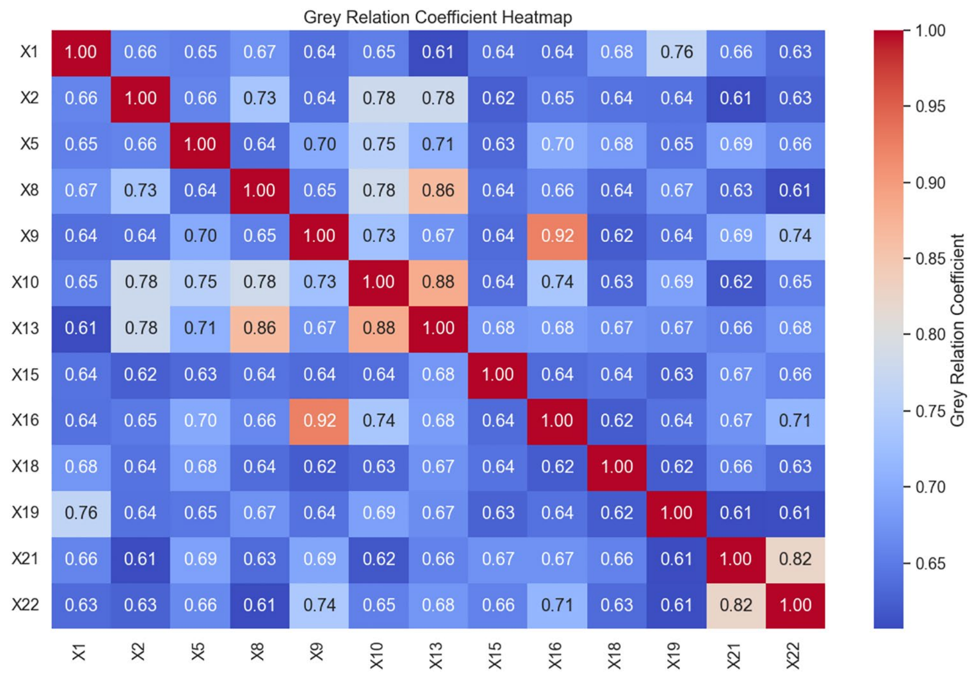

Fig. 1. Correlation matrix of various factors.

Model construction and analysis

Establishment of the gradient boosting decision tree model

The system modeling process in this study followed a logical framework of "data preprocessing—feature selection—model construction—model validation" (Fig. 2). First, the raw data were cleaned, outliers were processed, and all variables were standardized to ensure data quality and reduce bias. Secondly, the Grey Relational Analysis (GRA) method was employed to evaluate and rank the correlations between each feature variable and the set-level win rate, thereby optimizing the input feature structure and improving the model's signal-to-noise ratio. Based on the evaluation results, 13 highly correlated indicators were selected as the input variables for the model. Third, a prediction model was established using Gradient Boosting Decision Trees (GBDT), which included splitting the dataset into training and test sets, iterative model building, and hyperparameter optimization with performance evaluation. During hyperparameter tuning, a combination of grid search and K-fold cross-validation was used to determine the optimal parameter set, including learning rate, number of decision trees, and maximum tree depth, thereby enhancing the robustness and generalizability of the model. For performance evaluation, mean squared error (MSE), mean absolute error (MAE), coefficient of determination (R^2), and mean absolute percentage error (MAPE) were applied to assess model accuracy. Finally, an external validation was performed using a dataset independent of the training samples to further confirm the stability of the model.

1. Data Set Division

The performance of machine learning models is significantly influenced by the proportion of the data set division [17]. Selecting the optimal ratio of the training set to the test set is crucial for enhancing model performance. Commonly used ratios for dividing the training set and test set include 8:2, 7:3, 6:4, and 5:5[49], which can be determined based on the original data set. Random stratified sampling and cross-validation techniques were employed to divide the data set into training and test sets. The test set proportions were set at 20%, 30%, 40%, and 50%, respectively, and the mean squared error (MSE) values for different proportions were calculated (see Fig. 3). As shown in Fig. 3, when the ratio of the training set to the test set was 8:2, the MSE value was the smallest (0.004), indicating the smallest prediction error. Therefore, the data set was divided according to the 8:2 ratio.

2 Hyperparameter Tuning

After confirming the raw data, the dataset was split into training and test sets at a ratio of 8:2. The mean values of the 13 factors selected by Grey Relational Analysis (GRA) were used as input features, while the set win rate served as the output variable. A Gradient Boosting Regressor (GBR) function was employed to construct the GBDT-based prediction model for match-winning determinants in world women's volleyball.

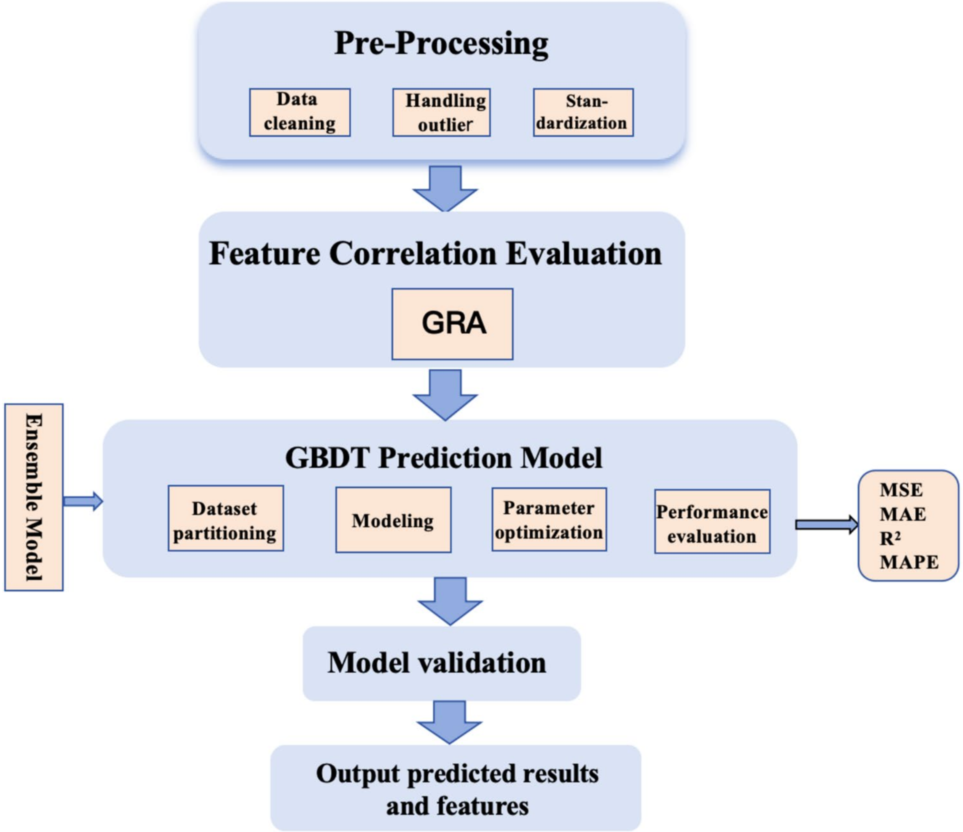

Fig. 2. System modeling flowchart.

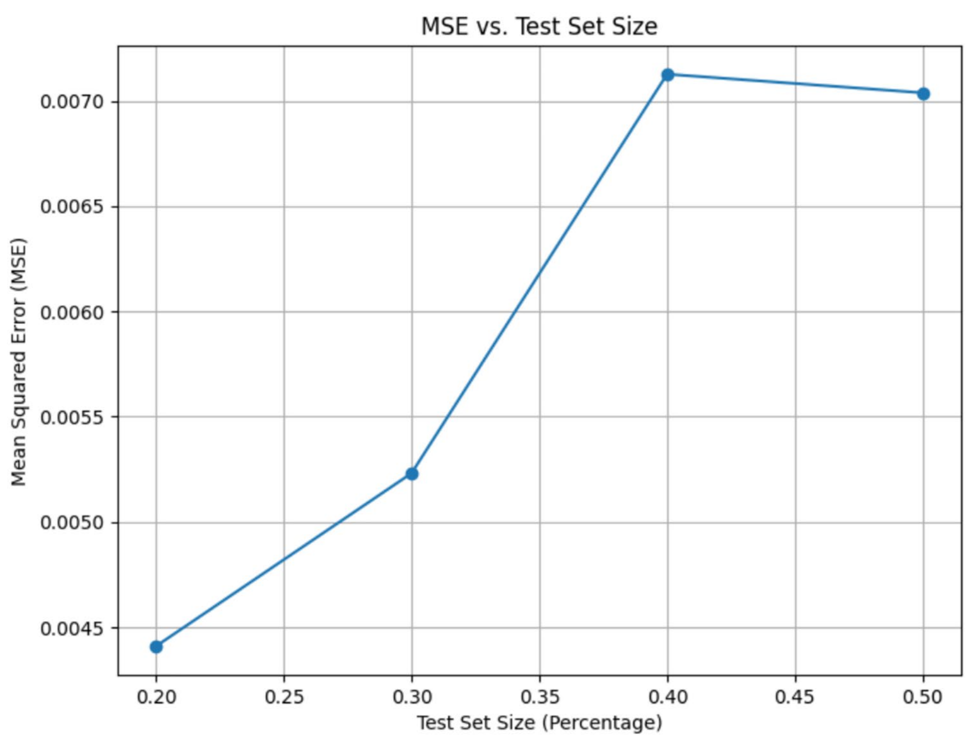

Fig. 3. Line graph of different data set divisions.

The parameters of the GBDT model, which are critical to its predictive accuracy and generalization ability, can be categorized into model parameters and hyperparameters. Model parameters are internal configuration variables generated automatically during the iterative learning process, whereas hyperparameters are external settings that must be manually specified and exert substantial influence on model performance. The hyperparameters most influential for GBDT include subsample, learning rate, number of estimators (n-estimators), and maximum depth (max-depth) [50] (see Table 3).

The learning rate controls the contribution of each tree, with a typical range between 0.01 and 0.2, and commonly adopted values such as 0.1, 0.05, and 0.01. In this study, the range was set with reference to existing literature and the characteristics of sports data, balancing convergence speed with model stability; excessively high values may lead to oscillations, whereas excessively low values slow convergence. The max-depth determines the depth of each decision tree, i.e., the number of splitting layers. Typical values range from 2 to 10: overly large values risk overfitting, while overly small values weaken model expressiveness. Common values include 2, 3, and 4. The n-estimators specifies the number of decision trees trained in the boosting process, usually ranging from 50 to 500. Too few trees restrict model complexity, while too many increase computational burden and the risk of overfitting; typical values include 50, 100, and 150[51]. The subsample is a regularization technique in GBDT, usually set at the default value of 0.8, whereby only a portion of samples are used in each iteration. This reduces reliance on the training data and effectively alleviates the risk of overfitting under small-sample conditions. The ranges of hyperparameters applied in this study are summarized in Table 3, providing a solid foundation for subsequent model optimization and performance evaluation.

There is a trade-off relationship between the aforementioned hyperparameters. Different datasets and problems require different combinations of hyperparameters, making hyperparameter tuning an essential practice[52]. Common methods include Random Search, Cross Validation (CV), Bayesian Optimization, and Grid Search[53]. Relevant literature indicates that a combination of Grid Search and Cross Validation is currently the most widely used approach[54].Grid Search is a hyperparameter tuning method used to find the best hyperparameter combination to optimize the performance of machine learning models. Cross Validation, specifically K-fold cross-validation, is a technique used to evaluate model performance by splitting the training data into different subsets for training and validation, which helps reduce the randomness in model performance evaluation[55]. The main advantages of combining Grid Search and Cross Validation are the comprehensiveness of parameter combinations, reproducibility of results, and automation of machine operations, leading to the identification of the optimal hyperparameter combination. Grid Search iterates through all possible hyperparameter combinations, with each combination used for training and cross-validation. During this process, MSE or RMSE is used to evaluate model performance. Therefore, this study employs Grid Search combined with Cross Validation for hyperparameter tuning of the model.

The hyperparameters n-estimators, learning-rate, and max-depth are tuned using Grid Search combined with Cross Validation. The value ranges for the hyperparameters are as follows: learning-rate is set to 0.01, 0.05, 0.1, and 0.2; n-estimators range from 50 to 500 in increments of 50; max-depth is set to 2, 3, 4, 5, 6, 7, 8, 9, and 10. The scoring is set to the mean relative error, with fivefold cross-validation applied. The first step involves tuning max-depth, with the system default set to 3, allowing for slight variations across different datasets (see Fig. 4). The results indicate that the MSE is minimized (0.013) when max-depth is set to 2. Thus, this study determines the optimal max-depth to be 2.

Upon determining the optimal max-depth, the next step is to identify the best values for n-estimators and learning-rate (see Fig. 5). In Fig. 5, the X-axis represents n-estimators, the Y-axis represents learning-rate, and the Z-axis represents the mean squared error (MSE) for performance evaluation. The X-axis shows that n-estimators start at 50 and increase incrementally, with performance peaking around 150 estimators. Beyond this point, performance may slightly decline, indicating that while increasing the number of decision trees enhances model complexity and computational cost, there is an optimal number of trees beyond which the model may overfit. The Y-axis depicts learning-rate values ranging from 0.01 to 1.0. A smaller learning rate results in slower learning because each tree contributes less to the overall model, while a larger learning rate accelerates learning but may cause instability. The performance curve indicates a lower MSE around a learning rate of 0.1. The Z-axis, representing negative MSE, shows that lower values indicate better model performance. In the three-dimensional plot, changes in X and Y values result in color shifts on the Z-axis, with deeper purple hues indicating better model performance.

From Fig. 5, it is evident that the optimal combination of hyperparameters occurs at specific MSE values. By comprehensively considering the information from the X, Y, and Z axes, the best-performing hyperparameters are identified as follows: max-depth of 2, n-estimators of 150, and learning-rate of 0.1. This combination corresponds to the lowest MSE, indicating the best model performance and generalization ability. Consequently,

Hyperparameter Name	Description	Search Range
n-estimators)	Determines the number of decision trees in the ensemble, affecting model complexity and predictive capability	50 ~ 500
learning-rate)	Controls the speed at which the model learns, impacting convergence speed and the risk of overfitting	0.01 ~ 0.2
subsample)	Randomly selects a portion of training samples for each iteration to construct the decision tree, reducing the risk of overfitting and enhancing model robustness	–
max-depth	Limits the maximum depth of the trees to prevent overfitting	2 ~ 10

Table 3. Hyperparameters of the GBDT Model.

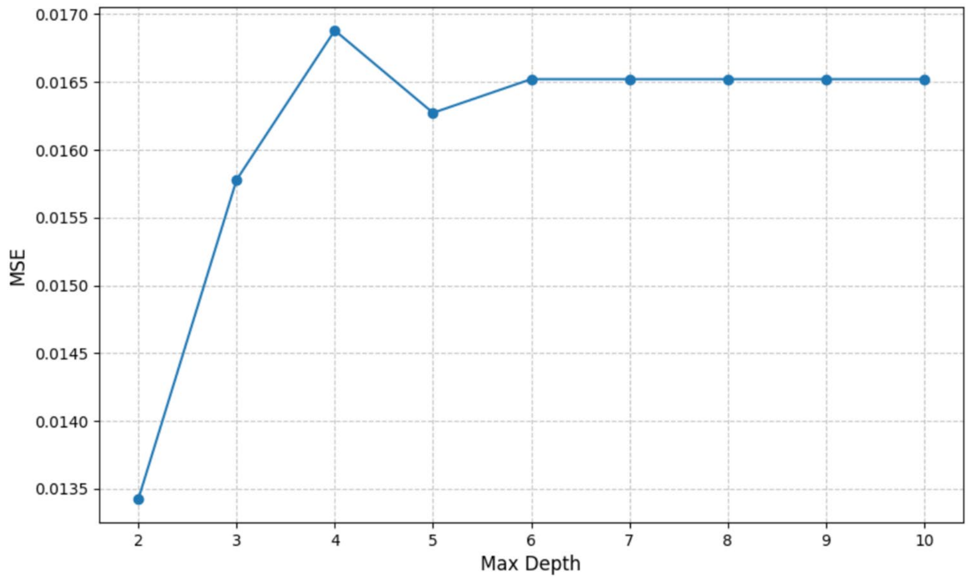

Fig. 4. Line graph for determining maximum depth.

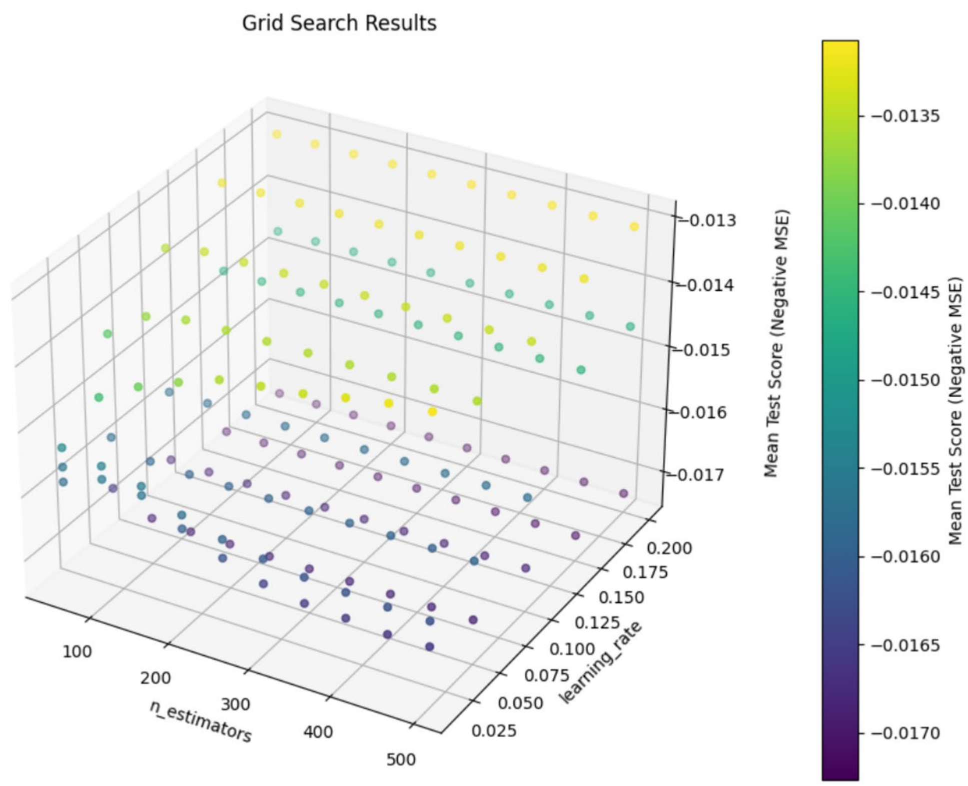

Fig. 5. 3D visualization of hyperparameter tuning performance.

this hyperparameter combination is used for constructing the model, ensuring optimal performance and robustness.

Analysis of gradient boosting decision tree model results

The evaluation metrics for the prediction accuracy of machine learning models primarily include Mean Squared Error (MSE), Mean Absolute Error (MAE), the Coefficient of Determination (R^2), and Mean Absolute Percentage Error (MAPE). The formulas for these metrics are as follows:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{9}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\backslash e_i\backslash = \frac{1}{n}\sum_{i=1}^{n}\backslash\,(\mathrm{y_i} - \overline{\mathrm{y_i}})\,\backslash \tag{10}$$

$$\text{R2} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(\mathrm{y_i} - \overline{\mathrm{y_i}})^2} \tag{11}$$

$$\text{MAPE} = \frac{100\%}{\mathrm{n}}\sum_{\mathrm{i=1}}^{\mathrm{n}}\backslash\frac{\mathrm{y_i}_{-}\hat{y}_i}{\mathrm{y_i}}\backslash \tag{12}$$

Lower MSE and MAE values indicate higher model accuracy and better generalization ability. The R^2value ranges from 0 to 1, with values closer to 1 indicating a better regression fit and higher accuracy. A lower MAPE value indicates higher model accuracy, with a value of 0 representing a perfect model and 100% representing the poorest accuracy.

The training and test sets were divided in an 8:2 ratio, with the test set containing 4 samples: specifically, Samples 1, 11, 12, and 15. A comparison of the predicted and actual values is shown in Table 4. The actual win rates were 77.27%, 54.76%, 52.50%, and 76.00%, with corresponding predicted values of 76.80%, 57.41%, 51.16%, and 84.43%. The model's predictions are relatively accurate. The performance metrics of the model are MSE = 0.002, MAE = 0.0322, R^2 = 0.8497, and MAPE = 4.77%, indicating good model performance and strong generalization ability. Therefore, this model can be used to predict set win rates in world women's volleyball matches.

Ranking of input feature importance in the world women's volleyball match winning prediction model
The feature importance in the Gradient Boosting Decision Tree (GBDT) model reflects the impact of different features on the target variable and the contribution rate of input features to the prediction of output features[56]. Re-ranking the input features helps analyze the factors influencing the output features. In this study, the average selection frequency of input features was used as a statistic to measure their importance. This method determines the contribution of each input feature to the prediction results by evaluating the frequency with which each feature is selected as a splitting node in the decision trees. Specifically, the more frequently a feature is selected during the decision-making process, the greater its impact on the target prediction. This frequency, after normalization, can be regarded as an indicator of the relative importance of the feature.

The feature importance of each indicator was ranked, and those with higher values were selected (see Fig. 6). The top features were spike success rate (X_{10}), block height (X_5), excellent defense rate (X_{22}), serve success rate (X_{16}), block success rate (X_{13}), serve score ratio (X_9), block score ratio (X_8), and height (X_2). Among these, the spike success rate and block height were identified as crucial determinants of match victories, with feature importance values of 0.4515 and 0.3040, respectively. The levels of these two indicators will directly affect the match outcome.

To further validate the robustness of the results, this study employed five-fold cross-validation combined with 200 repeated resampling to statistically analyze the distribution of feature importance and to calculate the median values with 95% confidence intervals (CIs). The results showed that the median feature importance of spike scoring rate (X_{10}) was 0.430 (95% CI [0.0797, 0.863]), and that of blocking height (X_5) was 0.231 (95% CI [0.041, 0.692]). Both were significantly higher than other indicators, and their confidence intervals did not cross zero, indicating that these two core features consistently maintained stable positive contributions under different sampling and modeling conditions, with clear statistical significance and robustness. In contrast, indicators such as excellent defense rate (X_{22}), block scoring rate (X_{13}), and proportion of block scores (X_8) showed certain contributions under specific conditions, but their confidence intervals were relatively wide, suggesting weaker robustness compared with the core indicators.

Sensitivity analysis was further applied to examine the degree to which the results respond to small perturbations in input variables, thereby assessing the robustness of model conclusions under random disturbances[57]. Based on the baseline model, 5% random noise was introduced to conduct the sensitivity test. The results indicated that the median feature importance of spike scoring rate (X_{10}) was 0.685 (95% CI [0.288, 0.867]), and that of blocking height (X_5) was 0.082 (95% CI [0.004, 0.238]). Both indicators consistently maintained positive contributions under perturbation, with their median values significantly higher than those

Sample	ID	Actual Value (%)	Predicted Value (%)
1	1	77.27	76.80
11	2	54.76	57.41
12	3	52.50	51.16
15	4	76.00	84.43

Table 4. Comparison of Predicted Values and Actual Values in the Test Set.

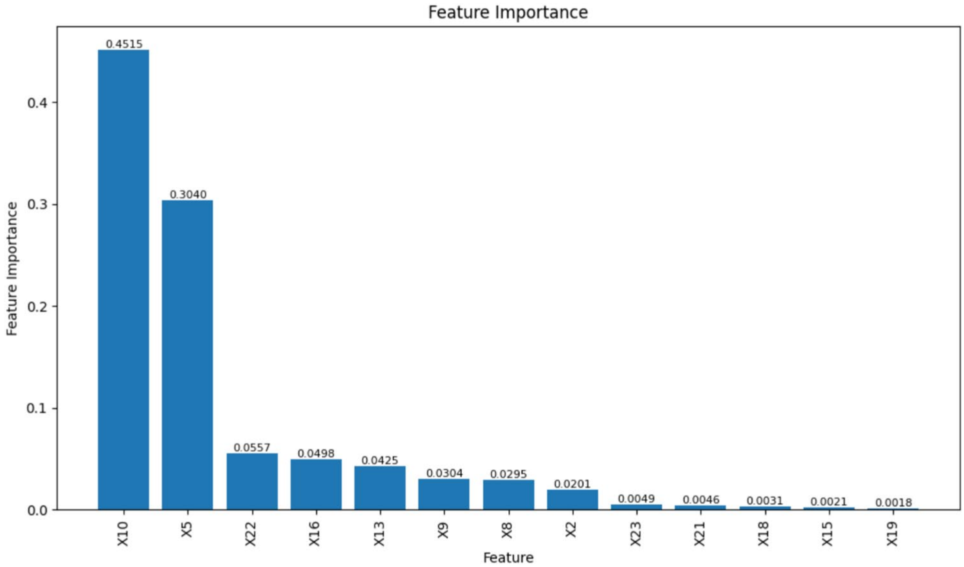

Fig. 6. Ranking of feature importance.

Country	X_{10}(%)	X_{13}(%)	X_5 (m)	X_{16}(%)	X_9(%)	X_{22}(%)	X_2(m)	X_{18}(%)	X_{19}(%)	X_8(%)	X_{21}(%)	X_{15}(%)	X_1 (y)	Win Rate(%)
SRB	50.78	24.77	2.87	4.70	5.96	60.77	1.85	82.02	7.65	14.25	91.35	41.22	27.98	80.49
BRA	44.81	24.23	2.81	4.58	5.79	58.83	1.84	86.64	9.94	16.04	89.43	37.70	28.15	69.00
USA	45.42	22.85	2.89	4.56	5.78	64.93	1.84	84.38	4.56	15.01	84.38	47.57	26.30	76.74
ITA	43.75	22.77	2.96	7.80	10.23	57.92	1.86	78.59	7.17	13.92	91.86	42.86	26.58	59.52

Table 5. Statistics for the Top Four Teams in the 2022 Women's Volleyball World Championship.

of other features, confirming their critical role across different sampling and modeling settings. In comparison, the importance distributions of other features exhibited wider intervals, reflecting lower stability. These findings underscore the central role of spiking and blocking in the mechanism of match victories, highlighting their robustness and statistical significance, thereby enhancing the credibility of the model conclusions and providing more reliable quantitative evidence for training design and tactical deployment.

In summary, through dual validation with confidence interval estimation and sensitivity analysis, this study confirms the core status of spike scoring rate (X_{10}) and blocking height (X_5) in determining victories in world women's volleyball. Both features consistently showed stable and significant positive contributions under varied sampling and noise-perturbation conditions, demonstrating statistical significance and methodological robustness. These results not only strengthen the credibility of the model conclusions but also provide persuasive quantitative evidence to support training organization, tactical planning, and post-match review.It should be noted that prior to model construction, this study introduced Grey Relational Analysis (GRA) to evaluate feature correlations and optimize structural composition, aiming to enhance the model's signal-to-noise ratio and robustness without losing essential information. GRA assesses the strength of association based on the relative variation trends between individual indicators and the win-rate sequence, whereas the subsequent Gradient Boosting Decision Tree (GBDT) model determines comprehensive weights through the marginal contributions of variables under multivariate conditions. Due to the differing analytical logics of the two methods, discrepancies emerged in the feature importance rankings: indicators such as X_{13} (block success rate), X_{16} (serve point rate), and X_9 (serve point proportion) ranked higher in the GRA analysis but demonstrated relatively lower contributions in the GBDT results. This divergence does not indicate methodological conflict; rather, it highlights the interpretive complementarity between the two approaches. GRA enhances the representativeness of feature selection and the rationality of the input structure, while GBDT reveals the true marginal effects of variables under multistage competitive conditions. Integrating the outcomes of both methods ensures the robustness of the model structure and empirically confirms the positive effect of GRA-based pre-screening in improving GBDT predictive accuracy, thereby achieving a unified optimization of feature selection and interpretability.

Model validation

To examine the applicability of the model, this study selected the top four teams from the 2022 Women's Volleyball World Championship (Serbia, Brazil, the United States, and Italy) as validation objects. Table 5 presents the average values of the 13 core indicators identified in the feature selection stage, which were used as input conditions for predicting set win rates. As shown in the table, among the previously confirmed important

Country	Predicted Win Rate (%)	Actual Win Rate (%)	Relative Error Rate (%)
SRB	75.60	80.49	6.08
BRA	70.25	69.00	1.81
USA	71.89	76.74	6.32
ITA	63.67	59.52	6.97

Table 6. Empirical Evaluation Results on the Model. Relative Error Rate = (Actual Win Rate—Predicted Win Rate) / Actual Win Rate.

features, both Serbia and the United States maintained high spike scoring rates (X_{10}), corresponding to their relatively high actual win rates (80.49% and 76.74%, respectively). Although Italy exhibited a clear advantage in blocking height (X_5), its spike scoring rate was comparatively lower, resulting in an overall win rate of only 59.52%, indicating that a single technical strength could not compensate for limited offensive efficiency. In contrast, Brazil demonstrated balanced performance across both indicators, with its actual win rate closely matching the predicted value, reflecting the stability of its overall technical-tactical structure.

The match-winning prediction model was used to predict the win rates of the top four teams in the 2022 Women's Volleyball World Championship. This model used 13 indicators selected through Grey Relational Analysis as input variables (detailed in Table 2). The model's three major hyperparameters, max-depth, n-estimators, and learning-rate, were set to their optimized values. The predicted win rates were output and compared with the actual win rates, with the results shown in Table 6.

The comparison between the predicted and actual values shows that the prediction deviation for Brazil was the smallest, with a relative error of 1.81%, followed by Serbia and the USA, with errors of 6.08% and 6.32%, respectively. Italy had the highest relative error at 6.97%. The average relative error was 5.30%. Additionally, the model's performance evaluation metrics include Mean Absolute Error (MAE = 0.0379), Mean Absolute Percentage Error (MAPE = 5.29%), Mean Squared Error (MSE = 0.0017), and the Coefficient of Determination ($R^2 = 0.743$). These metrics indicate that the model has good predictive capability and can effectively capture the patterns and trends in the data, demonstrating that the match-winning prediction model constructed using the GBDT algorithm can make accurate predictions.

Although this model was built based on data from the 2020 Olympic cycle, it also accurately predicted the win rates of the top four teams in the 2022 World Championship, showing excellent robustness and broad applicability. This indicates that the model can effectively learn from past data and apply this learning to predict future events. The strong ability of the GBDT algorithm to capture and utilize historical data patterns is demonstrated, providing an effective tool for predicting the outcomes of women's volleyball matches and offering valuable reference for constructing similar prediction models in other sports.

Discussion

"Efficient spiking" is the core to winning matches

As a skill-dominated net-separated confrontational sport, spiking is a crucial scoring method in volleyball. It can directly score points, dampen the opponent's morale, and exert significant pressure on their defense, thereby dismantling their blocking and defensive systems[58]. Statistics show that spike scoring accounts for 75%-85% of the total points in a match. Top-level teams worldwide possess excellent spiking abilities, characterized by powerful, high, and fast spikes typical of European and American teams. In contrast, Asian teams like China exhibit more varied and well-coordinated small-ball techniques, reflecting regional characteristics while also showcasing unique traits[59].

The Chinese team mainly relies on strong attacks from the 4th position (48%), with comparable attack rates from the 2nd and 3rd positions (less than 20%), and relatively few back-row attacks, indicating a less pronounced three-dimensional offensive power but diverse attack points and routes. The Serbian and Italian teams have balanced attack proportions from the 2nd, 3rd, and back-row positions (around 20%), with relatively fewer attacks from the 4th position (around 40%). They utilize low-arc passes and quick, broad-wing attacks in a coordinated team offensive system.The USA primarily focuses on attacks from the 4th position (around 50%), with fewer back-row attacks (7%) and balanced attacks from the 2nd and 3rd positions. Their offense is characterized by fast organization, quick front-row player movements, and a well-coordinated three-dimensional attack style. The Brazilian team shows moderate attacks from the 4th position (40%-45%), a significant proportion from the 2nd and 3rd positions (15%-20%), and fewer back-row attacks (10%-15%), utilizing low-arc passes and quick, broad-wing team attacks.

Regardless of the attack style, converting attacks into match points and enhancing offensive scoring efficiency in actual play is essential. This study selected spiking-related indicators such as spike height, spike score ratio, spike error rate, general spike rate, and spike scoring probability. Through initial screening via Grey Relational Analysis (GRA) and ranking the influencing factors in the Gradient Boosting Decision Tree (GBDT) match-winning prediction model, it was found that spike success rate has the most significant impact on winning matches, with a feature contribution value of 0.45.

The spike success rate refers to the percentage of successful spikes that directly score points out of the total number of spikes. It is an important indicator reflecting spiking efficiency, volleyball offensive capability, individual skill level, and the effectiveness of tactics during matches. The average spike success rate for teams during the 2020 Olympic cycle was 44.1%, and it was 44.83% during the 2018 World Championship, with the highest being the champion Serbia at 48.30%. The spike success rates of the top six teams (Serbia, Italy, China, Netherlands, USA, Japan) were nearly identical to their match rankings and set win rate rankings (except for a slight difference between the third-ranked China and second-ranked Italy in set win rate), as shown in Table 3.

Data from the 2019 World Cup showed an average spike success rate of 44.28%, with the highest being the champion China at 50% (the only team to exceed 50% during this Olympic cycle). A T-test showed a significant difference in spike success rates between China and other teams ($P < 0.05$). China averaged 15 spike points per set, the most among all participating teams, completing 11 matches with a total of 36 sets (the fewest among all participating teams). China's total number of spikes and spike points were the lowest among the top six teams, with 1080 and 540, respectively. Through the highest spike success rate, China won the World Cup with an undefeated record and a set win rate of 91.67%. The second-place USA had a set win rate of 75%, with a total of 1249 spikes and a spike success rate of 46.52%, scoring 581 points. The third-place Russia had a spike success rate of 46.3%, and the fourth-place Brazil had a rate of 42.9%. The rankings in match performance, set win rate, and spike success rate were consistent among the top four teams, further validating that the spike success rate is crucial to winning matches.

Since the Women's Volleyball World Cup has a different format compared to the World Championship and the Olympics, the 2019 World Cup featured 12 teams, with each team playing against all other teams once, totaling 11 matches per team. The ranking was determined by wins and then by points, with no elimination rounds or repeated matches. This format makes the match data more representative.

The performance of main attackers and opposite hitters, who bear the primary offensive responsibilities in matches, plays a crucial role in the team's overall spiking performance and match success. Studies have shown that efficient spiking not only requires strong attacks from excellent main attackers in the 4th position but also necessitates the collaborative support of opposite hitters, forming a coordinated spiking attack system[60]. Analyzing the performance of key attackers in matches, the ranking of the best attackers in the 2018 World Championship is led by Serbia's Tijana Boškovic (opposite) with a spike success rate of 53.66%; the second and third positions are held by Italy's Myriam Fatime Sylla (main attacker) and Paola Ogechi Egonu (opposite), with spike success rates of 49.65% and 48.85%, respectively. Fourth place is held by Netherlands' Lonneke Slöetjes (opposite) with 47.84%; fifth is Serbia's Brankica Mihajlovic (main attacker) with 47.28%; sixth is China's Zhu Ting (main attacker) with 45.5%; seventh is China's Gong Xiangyu (opposite) with 45.49%; and eighth is Netherlands' Anne Buijs (main attacker) with 40.2%.

Among the top eight players, they are the key main attackers and opposite hitters from the top four teams: Serbia (1st, 5th), Italy (2nd, 3rd), China (6th, 7th), and Netherlands (4th, 8th). The average spike success rates for the main attacker and opposite hitter pairs are 50.47% for Serbia, 49.25% for Italy, 45.5% for China, and 44.01% for the Netherlands. The average spike success rate aligns perfectly with match win rates and rankings, further demonstrating that spike success rate is a critical factor influencing match outcomes.

Following the 2019 World Cup, the rankings of the best spikers were announced. China's Zhu Ting (main attacker) ranked first with a spike success rate of 54.64%, with a total of only 280 spike attempts (12th place), scoring 153 direct points. Zhang Changning ranked second with a spike success rate of 48.26%, with a total of 201 attempts and 97 direct points. The 3rd to 8th positions were held by USA's Andrea Drews (opposite) with 46.67%, Russia's Kseniia Parubets (main attacker) with 45.38%, Dominican Republic's Brayelin Elizabeth Martinez (main attacker) with 45.36%, Russia's Irina Voronkova (opposite) with 45.29%, Russia's Nataliya Goncharova (main attacker) with 44.88%, and USA's Kelsey Robinson (main attacker) with 44.78%. The combined average spike success rates for the best main attackers and opposite hitters from each team were 51.45% for China, 45.73% for the USA, and 45.34% for Russia, consistent with their match win rates and rankings, showing clear linearity with the 2018 World Championship.

The data highlights the importance of spike success rate as a key indicator of offensive success, emphasizing the necessity of excellent individual skills and coordination among players. In practice, coaching teams should focus on improving players' spiking techniques, particularly the offensive power of main attackers in the 4th position and the supporting role of opposite hitters. Additionally, enhancing the coordination between main attackers and opposite hitters to form an effective offensive system is crucial. Coaches should also analyze opponents' defensive characteristics deeply and adjust tactics flexibly to maximize the scoring potential of main attackers and opposite hitters. Moreover, based on the characteristics of different teams, an offensive system that aligns with the team's unique strengths should be developed. For the Chinese women's volleyball team, enriching offensive strategies beyond the 4th position and increasing the proportion of back-row and 2nd/3rd position attacks would achieve a more three-dimensional offense.

"Blocking height" is key to controlling match offense and defense

Blocking is the first line of defense in volleyball and an aggressive form of defense, becoming an important means of proactive defense and scoring. Successful blocking can shut down the opponent's attack and directly score points, intercept their offensive rhythm, and alleviate the pressure on back-row defense, thus enabling counter-attacks[61]. This study selected initial indicators such as blocking height, blocking score ratio, blocking success rate, blocking error rate, and general blocking rate to analyze match-winning factors. Through initial screening via Grey Relational Analysis (GRA) and ranking the influencing factors in the Gradient Boosting Decision Tree (GBDT) match-winning prediction model, it was found that blocking height had the most significant impact on winning matches, with a feature importance of 0.304, followed by blocking success rate and blocking score ratio, with feature importances of 0.0425 and 0.0295, respectively.

Blocking height refers to the highest point at which a player can block the opponent's hit with their hands. It is the foundation for effective blocking[62]. Without ensuring sufficient blocking height, the defense at the net becomes ineffective and impacts the positioning of back-row defenders, leading to reduced overall defensive efficiency and inability to complete defensive counter-attacks. Research has shown that if the blocking height is between the minimum and median heights that can effectively intercept spikes, regardless of the speed, over 50%

of spikes will be non-threatening due to insufficient height. When the blocking height is between the median and maximum effective interception heights, at least 50% of spikes can be effectively intercepted as long as the speed meets the requirements. When the blocking height exceeds the maximum effective interception height, accurate judgment and movement can effectively intercept all spikes.

The average blocking height of the top six teams during the 2020 Olympic cycle's three major competitions was 2.92 m, with 2.93 m in the 2018 World Championship, 2.90 m in the 2019 World Cup, and 2.94 m in the 2020 Olympics. The 2018 World Championship champion, Serbia, had an average blocking height of 2.95 m, with the highest being China at 3.02 m. In this competition, the blocking heights were relatively even among teams, with China at the highest at 3.02 m and Japan the lowest at 2.86 m; the other four teams were around 2.94 m. In the 2019 World Cup, the average blocking height of the top six teams was 2.89 m, with China the highest at 3.02 m and the USA the second at 2.98 m. The blocking heights of the top six teams were consistent with their match rankings and set win rates. In the 2020 Olympics, the champion USA had the highest blocking height at 2.99 m, and the lowest was South Korea at 2.87 m, with other teams around 2.94 m, all reaching high levels.

The middle blocker is the main executor and initiator of blocking tasks in the team. While completing solo blocks, they need to coordinate with the opposite and main attackers to complete multi-person blocks. Measuring the blocking height of the middle blockers and the average blocking heights of the main attackers, middle blockers, and opposites (referred to as the three-position average blocking height) more accurately reflects a team's blocking height level and overall blocking performance[63]. In the 2018 World Championship, the average blocking height of middle blockers from the top six teams was 3.00 m, and the three-position average blocking height was 2.99 m. Serbia's middle blockers had an average blocking height of 3.02 m, with a three-position average of 3.01 m; Italy's were 3.02 m and 3.02 m, China's were 3.06 m and 3.09 m, the Netherlands' were 3.00 m and 2.98 m, the USA's were 3.00 m and 3.00 m, and Japan's were 2.90 m and 2.91 m. The rankings of blocking heights were consistent with match rankings and win rates. In the 2019 World Cup, the average blocking height of middle blockers was 2.96 m, and the three-position average was 2.95 m. The top six teams had the following blocking heights: China (3.11 m and 3.07 m), USA (2.97 m and 3.00 m), Russia (2.97 m and 2.96 m), Brazil (2.90 m and 2.90 m), Japan (2.89 m and 2.89 m), and South Korea (2.87 m and 2.88 m). The blocking heights and three-position averages were consistent with match rankings and win rates, demonstrating the impact of blocking height on match success.

In the 2020 Olympics, the average blocking height of middle blockers from all teams was 3.01 m, with a three-position average of 2.99 m. The USA had the highest averages at 3.03 m and 3.05 m, respectively. The other teams' blocking heights were also around 3.00 m. The match rankings and win rates of the top four teams were consistent with the blocking heights of the middle blockers and the three-position averages. Across the three major competitions, the relationship between blocking height and match win rates was evident, showing a strong correlation between blocking height and match success.

In the statistics of the best blocking heights, the average best blocking height in the three major competitions reached 3.10 m. The highest in the 2018 World Championship was Italy's opposite Paola Ogechi Egonu at 3.21 m; in 2019, it was China's Yan Ni at 3.15 m; and in 2020, it was Italy's Paola Ogechi Egonu again at 3.30 m. The highest blocking heights of the champion teams in the three competitions were around 3.14 m. Statistics show that the world's top main attackers can reach spike heights of over 3.2 m, indicating that only by ensuring effective blocking height can spikes be effectively interfered with.

Analyzing the average blocking heights of teams during this Olympic cycle, China's blocking height was the best, with an average blocking height of 3.09 m for middle blockers and 3.03 m for the three positions. Serbia and the USA had similar blocking levels, with average heights of around 3.01 m. Russia and the Netherlands had similar blocking heights of around 2.97 m. South Korea and Brazil were similar, at around 2.92 m, while Japan had the lowest at 2.89 m. Chinese players generally have taller stature, longer arm spans, and better static physical attributes, allowing for higher blocking heights. European and American teams, with certain height and arm span advantages and excellent physical qualities, achieve outstanding blocking heights through their height, arm span, and jumping ability. The Japanese team, due to its unique playing style focusing on excellent flexible defense rather than blocking, has relatively lower blocking heights.

Training suggestions and match strategies for the paris olympics

Targeted training suggestions to enhance match-winning capabilities

(1) Multidimensional Development of Spiking Abilities @@Strengthen the spiking abilities of key offensive players, focusing on improving the spiking techniques of main attackers and opposite hitters. This includes increasing spiking speed, power, and accuracy. Additionally, enhance the variability of spikes by incorporating feints and varying speeds to make the offense less predictable. To improve players' spiking success rates, training should emphasize simulating high-pressure match conditions by creating specific game scenarios and pressure conditions to mimic critical spike situations. This will help improve players' scoring abilities and mental resilience during crucial moments.@@For different positions, design specific spiking techniques and tactical application training, such as powerful back-row attacks for main attackers and quick diagonal attacks for opposite hitters. This comprehensive approach will enhance the team's offensive diversity and unpredictability. While strengthening individual skills, also focus on team collaboration and tactical understanding. Conduct internal scrimmages and group training sessions to improve mutual understanding and coordination among players, boosting overall offensive efficiency and tactical execution.@@ Furthermore, leverage video analysis and other technological tools to help players study and analyze the spiking techniques and tactical applications of top teams. This will inspire players' learning motivation and innovative thinking, thereby comprehensively enhancing their spiking success rates.

(2) Systematic Strategies for Improving Blocking Techniques @@To improve blocking height, in addition to conventional jump training, incorporate personalized strength and flexibility training programs based on players' physical conditions and technical characteristics to comprehensively enhance their physical fitness. The improvement of blocking techniques also requires precise guidance and immediate feedback from coaches. Utilize high-tech equipment to capture and analyze blocking movements, providing scientific training guidance and technical improvement suggestions for players.@@Enhancing blocking techniques is not just about increasing blocking height but also about improving players' timing, positioning, and coordinated blocking abilities. Training should include exercises for selecting the right blocking timing and adjusting positions. For instance, by setting up different attack routes and speeds, train players' predictive abilities and reaction speeds. Additionally, emphasize drills for coordinated multi-player blocking to improve overall blocking coordination and effectiveness, thereby enhancing the overall blocking performance.

Effective match decisions to enhance winning chances

(1) In-depth Analysis for Targeted Tactical Arrangements@@When formulating targeted tactics, the coaching team should thoroughly analyze the opponent's match videos, carefully studying their offensive habits, blocking setups, and defensive weaknesses to design tailored offensive combinations and formation strategies. Additionally, they should flexibly adjust the starting lineup and rotation strategies based on the team's strengths and current form, utilizing all tactical resources to maximize the team's overall advantage.@@In matches, the ability to adjust tactics is crucial for coaches. This requires keen observation and quick decision-making skills to promptly capture changes in the game, such as the opponent's strategic adjustments and players' condition changes. Coaches must be able to quickly make tactical adjustments or substitution decisions based on the actual situation. For example, if the opponent's main attacker shows signs of fatigue or instability in serve reception, adjusting the serve strategy to increase offensive pressure on them can effectively disrupt their offensive system.
(2) Enhancing the Application of Real-Time Data Analysis@@The role of data analysis is becoming increasingly prominent in modern volleyball matches. The coaching team should use real-time data analysis tools, such as match statistics software and intelligent analysis systems, to monitor and analyze key data during the match, including spike success rates, blocking success rates, and error rates. Real-time data feedback allows coaches to more accurately evaluate the effectiveness of match strategies, identify and correct problems promptly, and optimize offensive and defensive strategies.

Moreover, data analysis can help the coaching team gain deeper insights into players' performance trends and physical conditions, enabling more reasonable rotation arrangements during the match. For instance, by analyzing players' movement trajectories and physical exertion data, coaches can plan rest and rotation periods effectively, ensuring the team maintains a high level of competitive performance throughout the match.

Limitations and future directions

This study constructed a match-winning prediction model based on the Gradient Boosting Decision Tree (GBDT) algorithm using competition data from the top six teams in the three major world tournaments during the Tokyo 2020 Olympic cycle. The results showed that the model achieved a high level of accuracy and operational applicability, providing quantitative references for training arrangement and match strategy. However, several limitations remain in this study. In terms of sample representativeness, although the research sample covered the top six teams from the Olympic Games, the World Cup, and the World Championship, reflecting the competitive structure of elite-level volleyball, the applicability of the conclusions to different competition levels, regional tournaments, or youth competitions still needs further verification. Regarding data scale, although cross-validation and noise perturbation were employed to reduce the risk of overfitting, the model may still exhibit dependency on specific data distributions under limited sample conditions. Future research should systematically examine its stability and generalization ability using larger-scale, multi-level, and long-term datasets. In terms of modeling methodology, this study introduced Grey Relational Analysis (GRA) prior to GBDT modeling as a feature correlation evaluation method to optimize the feature structure and enhance model robustness. The empirical results indicated that the application of GRA contributed positively to improving model accuracy, but it may also lead to the omission of some potentially important features. Future studies could further compare different feature selection and dimensionality reduction strategies, such as LASSO regression, automated feature selection algorithms, and end-to-end modeling approaches, to validate the robustness and transferability of the results. Meanwhile, this study adopted only the GBDT algorithm for model construction. Although GBDT combines advantages in predictive performance and interpretability, it cannot fully represent the potential of other advanced algorithms. Future research could incorporate and compare XGBoost, LightGBM, and deep learning methods on an expanded dataset to further enhance predictive accuracy and structural stability. On this basis, future studies should integrate multimodal data, including technical–tactical indicators, physiological load, and psychological state, to explore the interaction mechanisms among different variables. By combining real-time data acquisition with dynamic modeling techniques, an adaptive and comprehensive analytical framework can be established, achieving greater completeness, robustness, and transferability. Such a framework would provide higher-level scientific support for optimizing training and in-game decision-making in competitive volleyball. The research framework proposed in this study could also be extended to other sports, such as basketball and football, offering a data-driven path for training evaluation and tactical decision-making across different disciplines.

Conclusion

(1) The match-winning prediction model constructed using the Gradient Boosting Decision Tree (GBDT) algorithm has parameter values as follows: learning rate (learning-rate) is 0.1, number of trees (n-estimators) is 150, and maximum depth of the tree model (max-depth) is 2. The accuracy metrics for the test set are as follows: MSE = 0.002, MAE = 0.0322, R^2 = 0.8497, and MAPE = 4.77%. The model exhibits high accuracy and can be used for predicting match win rates.
(2) The key indicators selected by the GBDT algorithm that are highly correlated with match-winning include spike success rate, blocking height, excellent defense rate, serve success rate, blocking success rate, serve score ratio, and blocking score ratio. Among these, spike success rate and blocking height have the most significant impact on match-winning, with feature importance values of 0.45 and 0.30, respectively. Therefore, training and match strategies should focus on improving spike success rate and blocking height to increase the probability of winning.
(3) This study validated the effectiveness of the GBDT prediction model and identified the roles of key technical–tactical indicators in determining match outcomes, providing empirical support for win-rate prediction and training decision-making. Future research should further examine the model's robustness and generalization ability using larger-scale, multi-level, and long-term datasets, while integrating multimodal data to deepen comparative studies of multiple algorithms and model fusion. By establishing a more adaptive and generalizable intelligent analytical framework, this line of research can offer systematic and sustainable data support for scientific training and match decision-making in volleyball as well as other competitive sports.

Data availability

The dataset analyzed in this study was derived from publicly available official match statistics published by the Fédération Internationale de Volleyball (FIVB). Specifically, data were collected from the FIVB official website, covering the top six teams in three major international tournaments during the 2020 Olympic cycle: the 2018 FIVB Volleyball Women's World Championship, the 2019 FIVB Women's World Cup, and the 2020 Tokyo Olympic Games. The dataset includes 27 technical and tactical performance indicators and set-based win ratios. All data supporting the findings of this study are accessible via the FIVB official website: https://www.fivb.com.
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Author contributions

M. designed the study, conducted data preprocessing, performed the modeling and statistical analysis, and drafted the main manuscript text. J. contributed to the theoretical framework design, assisted with interpretation of results, and revised and polished the manuscript. Both authors reviewed and approved the final version of the manuscript.

Competing interests

The authors declare no competing interests.

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